S3.1 The periodic table: Classification of elements

Group 1 and 17 elements

Learning outcomes

By the end of this section you should be able to:

- Describe the trend in metallic and non-metallic behaviour in the periodic table.
- Describe and explain the reactions of group 1 metals with water.
- Describe and explain the reactions of group 17 elements with halide ions.

In section S3.1.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-the-periodic-table-id-43466/) we looked at metals, metalloids and non-metals along with some of their properties. **Table 1** shows images and properties of aluminium, silicon and sulfur, all elements in period 3. Using **Table 1** how would you classify the type of each element (metal, metalloid, non-metal)? What clues did you use to classify them?

Table 1. Images and properties of aluminium, silicon an

	Aluminium	Silicon	
Image	 Credit: MirageC, Getty Images (https://www.gettyimages.com/detail/photo/crumpled-aluminium-foil-roll-royalty-free-image/1192681753)	 Credit: Oliver Helbig, Getty Images (https://www.gettyimages.com/detail/photo/high-purity-polycrystalline-silicon-from-germany-royalty-free-image/1395000196)	(s

		Aluminium	Silicon
Properties	Melting point (°C)	660.3	1414
	Conductivity	High	Moderate

Did you correctly identify aluminium as a metal, silicon as a metalloid and sulfur as a non-metal? While we as humans like to place elements into distinct categories, the reality is these metallic and non-metallic characteristics are a spectrum. How might we describe the trend of metallic or non-metallic character of an element as we travel down a group or across a period? How might you expect the metallic or non-metallic behaviour of an element to correlate with ionisation energy? What about electron affinity?

Metallic character

Metallic character is the tendency of an element to lose electrons to form positive ions. What quantity from section S3.1.3b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/periodicity-of-ionisation-energy-electron-affinity-id-44693/) would provide an indicator for the tendency of electron loss? Recall that ionisation energy is the energy required to remove the outermost electron from a neutral atom. The lower the ionisation energy, the greater the tendency that element has for electron loss and therefore the greater metallic character. This means that elements lower down to the left of the periodic table have the greatest metallic character. Which element do you think would have the most metallic character out of Mg, K and Ca? Check the box below to confirm your answer.

Interactive 1. Metallic Character.

More information for interactive 1

Non-metallic character

Non-metallic character is the tendency of an element to gain electrons to form negative ions. What quantity from section S3.1.3b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/periodicity-of-ionisation-energy-electron-affinity-id-44693/) would provide an indicator for the tendency of electron gain? Recall that electron affinity is the energy change that occurs when an element gains an electron. The higher the electron affinity, the greater the tendency that element has for electron gain and therefore the greater non-metallic character. This means that elements higher up to the right of the periodic table have the greatest non-metallic character. Which element do you think would have the most non-metallic character out of F, S and Cl? Check the box below to confirm your answer.

Interactive 2. Non-metallic Character.

More information for interactive 2

Figure 1 labels the trend in metallic and non-metallic character in the periodic table.

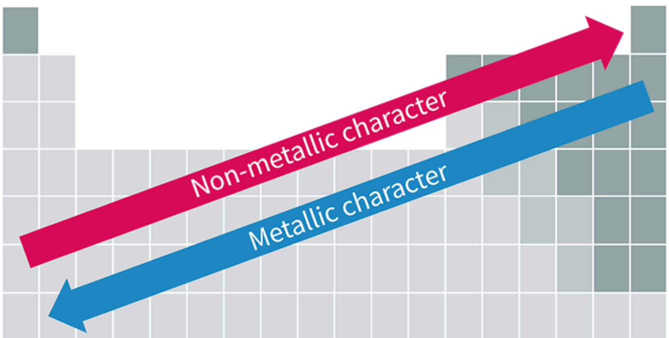


Figure 1. Trend in metallic and non-metallic character.

More information for figure 1

Study skills

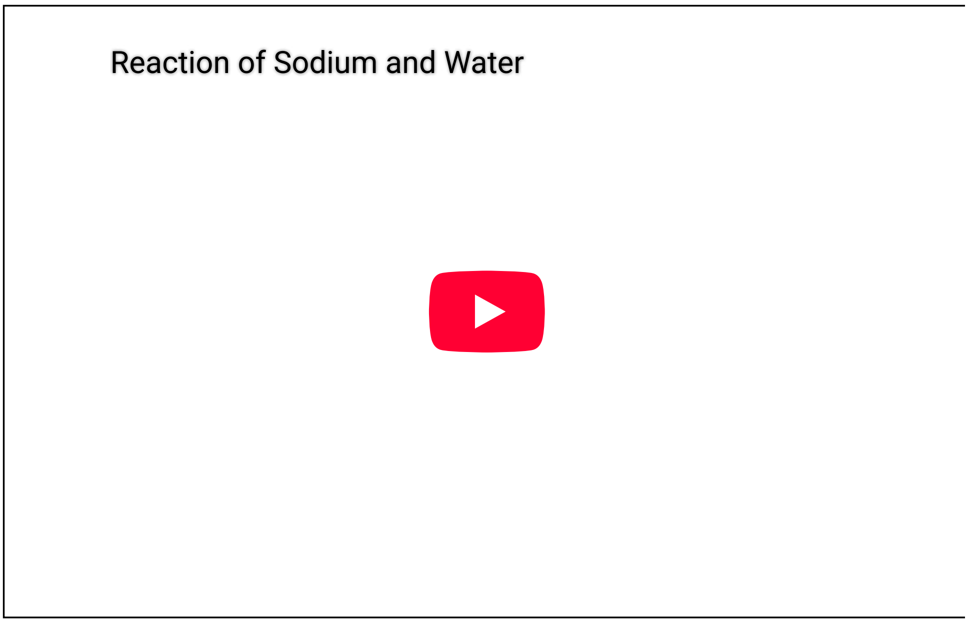
We will be using metallic and non-metallic character to help explain the chemical reactivity of group 1 metals with water and group 17 elements with halide ions. In these examples we will be looking at chemical reactions in terms of electron transfer (electrons transferred from one species to the other). This type of reaction will be the most familiar to you at this time, particularly after looking at the formation of ionic compounds in section S2.1.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/). It is important to note that there are other mechanisms for chemical reactions that do not involve electron transfer. We will look at these in the R3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/) subtopics.

Reactions of group 1 metals with water

While we often hear of group one metals such as sodium and potassium in foods, we most commonly encounter these elements as ions (Na^+ and K^+) where they have already lost their outer valence shell electron. Why is it uncommon to find these elements as pure metals? Why don't we see metallic solids made of group 1 metals as building structures or making up a surface in our kitchen?

The reason we cannot use group 1 metals for solid materials is that they are highly reactive with substances such as water. Since water is in our atmosphere it would not be possible to create solid materials out of these metals without a reaction occurring. This is why these substances are typically stored in oil to prevent reactivity with water.

Why are group 1 metals highly reactive with water? What do you think is occurring? What do you think the trend in reactivity down group 1 would be with water? **Video 1a** and **Video 1b** show the reactivity of the group 1 metals sodium and potassium with water. Watch each of them, record your observations and compare the reactivity of sodium and potassium with water. Which substance was the most reactive? Which was the least? Can you think of an explanation for why one would be more reactive than the other? Can you predict any of the products formed in this chemical reaction?



Video 1a. Reaction of sodium and water.

Video 1b. Reaction of potassium and water.

Practical skills

Tools 1: Experimental techniques — Addressing safety

The reaction of group 1 metals with water poses significant safety concerns. Notice in the clips in **Video 1** only a very small piece of the metal was used in the container of water. This is to minimise the risks to the observer.

Handling of group 1 metals should only ever be done by an experienced adult with a chemistry background. Many research chemists only work with large quantities of group 1 metals in an inert environment such as a glovebox where all atmospheric air has been replaced with an inert gas like argon. These metals should never be left out in the open air and should always be stored in oil. It is important that any tools used to handle the metal pieces be completely dry to avoid any reaction with the metal in the storage container.

The group 1 metals all react similarly with water. The reaction for sodium with water is:

$$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$$

You may have noticed bubbling in the videos showing these reactions. These bubbles produced show the production of a gas, in this case hydrogen gas. Since hydrogen is flammable it can ignite, you would have observed this in the reaction of potassium with water.

The other product NaOH is the formation of a compound containing hydroxide. You might recall from section S2.1.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/) that hydroxide is the polyatomic ion, OH^-. This means an ionic compound is formed from Na^+ and OH^- that is soluble in water, hence the aqueous state. Compounds containing hydroxides form basic solutions with a high pH. This is why you can see the presence of a pink colour forming in the videos as an acid-base indicator was used to show this change. You will learn more about the pH scale and bases in section R3.1.4-5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/calculations-involving-ph-and-[h+]-id-44702/).

Concept

The reactivity of group 1 metals with water is one of the first mentions of chemical reactions so far. As you might expect, reactivity is a very important concept in chemistry. In chemistry chemical equations are symbolic representation of chemical reactions that show reactants (starting materials) on the left-hand side and products (what is formed) on the right-hand side, separated by an arrow. Coefficients are used to represent the relative amounts of each substance. These can be deduced by ensuring the total number of atoms of each element are the same on the reactants side as the products side.

In order to form the ionic hydroxide compound, the originally neutral metal needed to lose its valence electron to water. We can evaluate the ease at which a metal loses its outer valence electron by comparing the values of first ionisation energy. **Table 2** shows the reaction, observations and first ionisation energies for Li, Na and K.

Table 2. Reactions, observations and first ionisation energies for Li, Na and K reacting with that contains phenolphthalein indicator.

Element	First ionisation energy (kJ mol^{-1})	Reaction with water	Qualitative observations f reaction with w
Li	520	$2Li(s) + 2H_2O(l) \rightarrow 2LiOH(aq) + H_2(g)$	Solid floats a moves slowly o surface, gen fizzing. Water b to turn pink du reaction.
Na	496	$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$	Solid melts to c a molten ball floats and mc more quickly o surface, fizzi Water begins t pink during rea
K	419	$2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$	Solid melts to c a molten ball

Element	First ionisation energy (kJ mol^{-1})	Reaction with water	Qualitative observations f reaction with w
			quickly ignites produces a pu flame. Water b to turn pink du reaction.

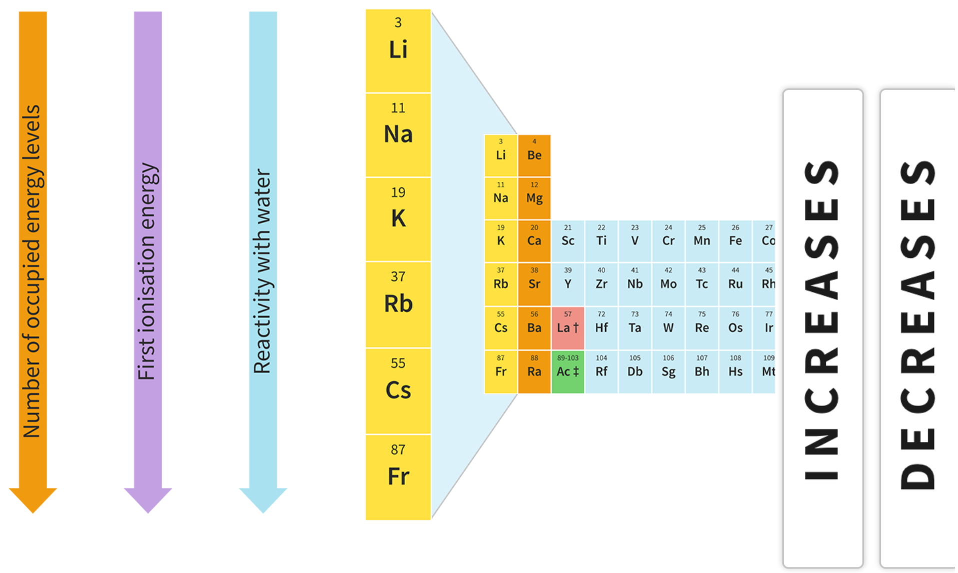
General trend of reactivity for group 1 metals with water

- Increasing reactivity going down group 1.
- Going down group 1, elements have lower first ionisation energies due to the presence of an additional energy level.
- Easier for with lower ionisation energy to transfer the outer valence electron to water increasing reactivity with water.

How would you predict the reactivities of Rb and Cs would compare with Li, Na and K? Watch the video to confirm your predictions.

Video 2. Reactivity of the group 1 metals with water.
Credit: Twenty47studio, Getty Images (https://www.gettyimages.com/detail/photo/sulfur-mining-at-mount-kawah-ijen-crater-in-east-royalty-free-image/1319947829)

Try **Interactive 3** as a drag and drop activity to review your knowledge of the general trend for number of occupied energy levels, first ionisation energy and reactivity with water for group 1 metals.



Check

Interactive 3. Test your knowledge.

More information for interactive 3

Reactions of group 17 elements with halide ions

The halogens are non-metallic elements in group 17 that exist as diatomic molecules. These elements are highly reactive with many substances since they have a tendency to gain electrons. We can quantify this tendency to gain electrons by looking at the values for electron affinity for these elements. Recall from section S3.1.3b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/periodicity-of-ionisation-energy-electron-affinity-id-44693/) that the general trend for electron affinity is that it decreases down a group. What do you think this means for reactivity? How do you think the general trend for reactivity will be for group 17 elements when they react in a manner to gain an electron?

A common type of reaction that the halogens readily undergo is called a displacement reaction. This is where part of a reactant replaces or displaces part of another reactant. These reactions are common to occur between halogens and halide salts. An example of this reaction is Cl_2 with KBr:

$$Cl_2(aq) + 2KBr(aq) \rightarrow Br_2(aq) + 2KCl(aq)$$

In this reaction, Cl_2 is the diatomic halogen and KBr is the ionic compound made up of the metal ion K^+ and a non-metal halide ion Br^-. In this type of displacement reaction, the extra electron that the non-metal ion Br^- has is transferred to the halogen to form the Cl^- ion. This results in each halogen atom in Cl_2, gaining an electron and each halide ion, Br^-, losing an electron.

Table 3 shows a comparison of the reactions involving chlorine and bromine along with their electron affinity values. In the first example, chlorine is reacting with sodium bromide. Since chlorine has a higher electron affinity than bromine, chlorine gains the extra electron from the bromide ion to form the chloride ion (Cl^-) and bromine (Br_2). Since sodium remains as the Na^+ ion throughout it is often referred to as a spectator ion and can be omitted from the reaction.

Notice how there is no reaction between Br_2 and NaCl. This is because bromine has a lower electron affinity than chlorine and therefore unable to gain an electron from the chloride ion. Only one combination of halogen and halide ions can result in a reaction, otherwise, the products would be able to react to form reactants over and over again (for example, $Cl_2(aq)$ will react with NaBr(aq) but $Br_2(aq)$ will not react with NaCl(aq)).

Table 3. Electron affinity and reactions for chlorine and bromine.

Halogen	Electron affinity ($kJ\ mol^{-1}$)	Reaction
Cl_2	—349	$Cl_2(aq) + 2NaBr(aq) \rightarrow 2NaCl(aq) + Br_2(aq)$ or $Cl_2(aq) + 2Br^-(aq) \rightarrow 2Cl^-(aq) + Br_2(aq)$
Br_2	—325	$Br_2(aq) + NaCl(aq) \rightarrow$ no reaction

Use **Video 3** to make observations of the reactions of aqueous solutions of Cl_2, Br_2 and I_2 with aqueous solutions of KCl, KBr and KI. Identify which combination of halogens and halides will lead to a reaction using this video.

Video 3. Reactivity of halogens with halide ions.

Worked example 1

State the reactions of aqueous solutions of Cl_2, Br_2 and I_2 as they react with KCl, KBr, KI. Explain why some combinations have no reaction.

Halogen (Cl_2, Br_2, I_2)	Reactions with KCl, KBr, and KI — Displacemen
Cl_2	Cl_2 + KCl → no reaction (no element to displace) Cl_2 + 2KBr → 2KCl + Br_2 (chlorine has a higher e bromine) Cl_2 + 2KI → 2KCl + I_2 (chlorine has a higher elec
Br_2	Br_2 + KCl → no reaction (bromine has a lower ele chlorine) Br_2 + KBr → no reaction (no element to displace) Br_2 + 2KI → 2KBr + I_2 (bromine has a higher elec
I_2	I_2 + KCl → no reaction (iodine has a lower electro I_2 + KBr → no reaction (iodine has a lower electro I_2 + KI → no reaction (no element to displace)

Try Interactive 4 a drag and drop activity to review your knowledge of the general trend for number of occupied energy levels, electron affinity and reactivity of the halogens with halide ions.

Interactive 4. The general trend of the number of occupied energy levels, electron affinity and reactivity of the halogens with halide ions.

More information for interactive 4

Activity

- **IB learner profile attribu**te: Inquirer
- **Approaches to learning:** Thinking skills — Providing a reasoned argument to support conclusions
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual first followed by paired discussion

Identifying unknown substances

You find six substances and suspect they might be the metals Li, Na, K, and aqueous solutions of Cl_2, Br_2 and I_2. You make observations of the substances and their reactions and record your observations in **Table 4.**

Table 4. Observations of unknown substances and the

Substance	Appearance	
AA	Stored in oil and very soft solid.	Vigorous r a
BB	Aqueous solution light orange-brown in colour.	Reaction o
CC	Stored in oil and soft solid.	Molten ba s
DD	Aqueous solution dark brown in colour.	No reac aque
EE	Aqueous solution very pale green in colour almost colourless.	Reaction o s
FF	Stored in oil. Firm.	Solid float

Individually, complete the following:

- Identify the six substances AA, BB, CC, DD, EE, and FF.
- For each substance explain how you came to your conclusion using both evidence and reasoning.
- Write out a chemical equation to represent the reaction described for each of the group 1 metal substances.

With a partner:

- Compare your identities of AA, BB, CC, DD, EE and FF. For any you and your partner do not agree on discuss your reasoning with one another and come to a consensus about the identity. Check in with your teacher.
- Discuss explanations for the trends in reactivity for:
  - Group 1 with water.
  - Group 17 with halide ions.
- Ensure your explanations include the following:
  - Description of general trend.
  - A quantitative property that supports that trend.
  - What is occurring in the reaction that connects to that quantitative property.
  - Discussion of the strength of the interaction between the nucleus and outer valence shell.